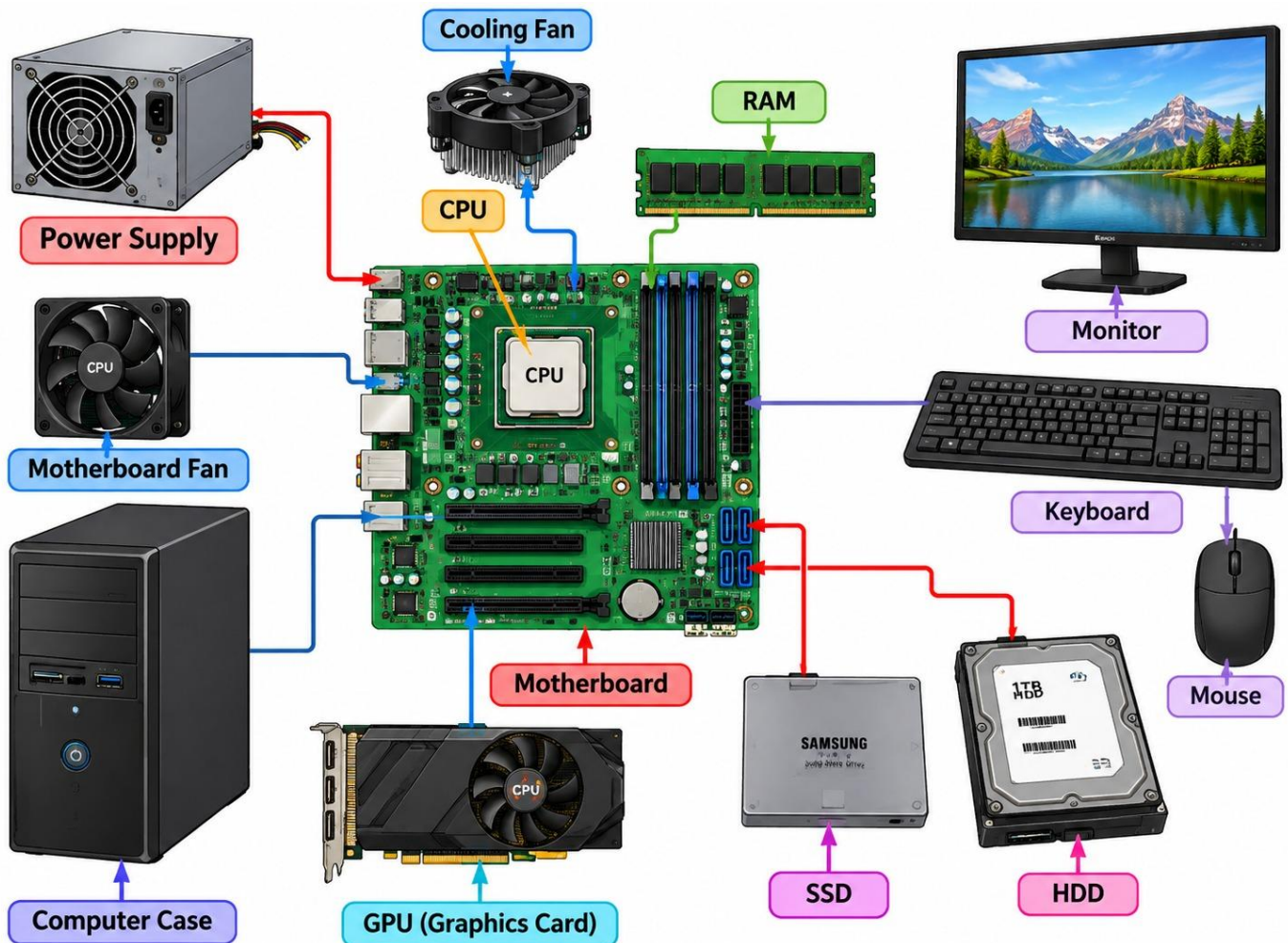


Overview of Hardware Components

1. Labeled Diagram of a Desktop PC



The motherboard connects the major components. The CPU processes instructions, RAM temporarily stores data being used, the GPU handles graphics, and HDD/SSD provide long-term storage.

Overview of Hardware Components

2. HDD vs SSD Comparison

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed (Read/Write)	Slower (mechanical disk rotation)	Significantly Faster (flash memory)
Durability & Shock Resistance	Lower (moving parts susceptible to physical shock)	Higher (no moving parts, more robust)
Typical Use Cases	Bulk storage, NAS, budget systems, archival data	OS drives, gaming PCs, laptops, high-performance workstations
Form Factor	Larger (typically 3.5" or 2.5")	Smaller (2.5", mSATA, or M.2)
Noise Level	Audible (spinning platters, seek arm)	Silent
Cost per Gigabyte	Lower	Higher

Overview of Hardware Components

3. Roles of CPU, RAM, and GPU in a Game

CPU: The CPU (Central Processing Unit) handles game logic, calculations, physics, and instructions from the operating system and game. A capable CPU helps maintain smooth gameplay and supports other components.

RAM: RAM (Random Access Memory) temporarily holds game data and programs that are currently being used. Enough RAM helps the game load assets smoothly and reduces slowdowns caused by excessive data swapping.

GPU: The GPU (Graphics Processing Unit) renders the game's images, textures, lighting, and visual effects. A stronger GPU can provide higher graphics settings and smoother frame rates when the rest of the system is balanced.

4. Hardware Upgrades for Running the Latest FIFA Smoothly

GPU (Graphics Card): Consider upgrading the GPU first if the current graphics card cannot meet the game's recommended requirements. A stronger GPU improves rendering performance, frame rates, resolution, and graphics quality.

CPU (Processor): Upgrade the CPU if it is older or has insufficient processing performance. A capable CPU helps with game logic, physics, background tasks, and maintaining stable frame rates.

RAM: Increasing RAM can help if the PC has too little memory or if many applications are running in the background. It gives the game more working memory and can reduce stuttering.

SSD: Installing the game on an SSD can significantly improve loading times compared with an HDD. An SSD is especially useful for faster booting, game launches, and level/menu loading.

Power Supply Unit (PSU): If a more powerful GPU or CPU is installed, check that the PSU can safely provide enough power. A suitable PSU also needs the correct connectors and adequate quality for the upgraded hardware.